

Assignment 2 [written]

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1 Attention exploration

(a)

- i By the definition of α_i in equation(2), all α_i are 1) non-negative, 2) softmax-normalized and 3) $\sum_{i=1}^n \alpha_i = 1$. These properties match the definition of a categorical probability distribution.
- ii When the dot product $k_j^\top q$ is much larger than $k_i^\top q$ for all $i \neq j$, it is. It needs the k_j is much closer to the q than other keys.
- iii Since $\alpha_j \approx 1$ and $\alpha_i \approx 0$ for $i \neq j$, the attention output $c = \sum_{i=1}^n \alpha_i v_i \approx \alpha_j v_j \approx v_j$.
- iv This means the "selective focus" of attention: when the query q (e.g., a word in the target sentence) has a strong matching relationship with a specific key k_j (e.g., a word in the source sentence), the model will "focus" almost all attention on the corresponding value v_j and directly "copy" this value to the output c .

(b)

- i - Express $v_a = \sum_{j=1}^m c_j a_j$ (where $c_j = a_j^\top v_a$ by orthonormality) and $v_b = \sum_{k=1}^p d_k b_k$ (where $d_k = b_k^\top v_b$).
 - Since $s = v_a + v_b = \sum_{j=1}^m c_j a_j + \sum_{k=1}^p d_k b_k$.
 - For $Ms = v_a$, means extract v_a from $s = v_a + v_b$, and we know $a_j^\top b_k = 0$ for all j, k .
- Define matrix M with rows $a_1^\top, a_2^\top, \dots, a_m^\top$:

$$M = \begin{bmatrix} a_1^\top \\ a_2^\top \\ \vdots \\ a_m^\top \end{bmatrix}$$

Compute Ms :

$$Ms = \begin{bmatrix} a_1^\top(v_a + v_b) \\ a_2^\top(v_a + v_b) \\ \vdots \\ a_m^\top(v_a + v_b) \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix} = \sum_{j=1}^m c_j a_j = v_a$$

- ii $q = \lambda(k_a + k_b)$, where $\exp(\lambda) \gg (n-2)$
 - Given all key vectors are orthonormal (i.e., $k_i^\top k_j = 0$ for $i \neq j$ and $\|k_i\| = 1$), if we get $c = \alpha_a v_a + \alpha_b v_b + \dots \approx \frac{1}{2}(v_a + v_b)$, we need $\alpha_a \approx \alpha_b \approx \frac{1}{2}$ and $\alpha_i \approx 0$ for $i \notin \{a, b\}$.
 - That means $k_a^\top q \approx k_b^\top q \gg k_i^\top q$.
 - Just like the previous question, When the dot product $k_j^\top q$ is much larger than $k_i^\top q$ for all $i \neq j$, it is. It needs the k_j is much closer to the q than other keys. It needs $q = \lambda(k_a + k_b)$.
- Justify this, let $q = 5(k_a + k_b)$. Then:

$$k_a^\top q = 5(k_a^\top k_a + k_a^\top k_b) = (5 + 0) = 5$$

$$k_b^\top q = 5(k_b^\top k_a + k_b^\top k_b) = (0 + 5) = 5$$

$$k_i^\top q = 5(k_i^\top k_a + k_i^\top k_b) = (0 + 0) = 0$$

Thus:

$$\alpha_a = \alpha_b = \frac{\exp(5)}{\exp(5) + \exp(5) + \sum_{i \notin \{a,b\}} \exp(0)}$$

Here we can see the denominator consists of three parts, to satisfy the condition requires $\exp(5) \gg n - 2$, which means the restrictions on λ .

Hence, $q = 5(k_a + k_b)$ satisfies the condition, and we can extend to any λ meeting the requirement.

(c)

i Since $k_i \sim \mathcal{N}(\mu_i, \alpha I)$ and α is vanishingly small, $k_i \approx \mu_i$. And we also know that all the mean vectors are perpendicular.

- To satisfy the condition, just like the previous question: $q = \lambda(\mu_a + \mu_b)$, where $\exp(\lambda) \gg (n - 2)$, now we design one example of them as $q = 10(\mu_a + \mu_b)$.

ii For the query $q = 10(\mu_a + \mu_b)$ and the new k_a :

Define δ to quantify the deviation of k_a from its mean direction μ_a , and analyze the behavior of c .

- Given $k_a \sim \mathcal{N}(\mu_a, \Sigma_a) = \mathcal{N}(\mu_a, \alpha I + \frac{1}{2}\mu_a\mu_a^\top)$, The projection of k_a onto μ_a (a key component for $k_a^\top q$) is defined as $k_a^\top \mu_a = 1 + \delta$, and δ is the random fluctuation around this mean.

- By the properties of normal distributions, $\delta \sim \mathcal{N}(0, 0.5)$ (zero-mean normal distribution with variance 0.5), meaning δ is equally likely to be positive or negative.

- Due to the orthogonality of μ_a and μ_b ($\mu_a^\top \mu_b = 0$) and small α , $k_a^\top \mu_b \approx 0$. The dot product $k_a^\top q$ simplifies to:

$$k_a^\top q = k_a^\top \cdot 10(\mu_a + \mu_b) = 10(k_a^\top \mu_a + k_a^\top \mu_b) \approx 10(1 + \delta) = 10 + 10\delta.$$

- In contrast, $k_b \sim \mathcal{N}(\mu_b, \alpha I)$ (stable, no extra variance), so $k_b^\top q \approx 10$ (constant), and $\exp(k_b^\top q) \approx \exp(10) \approx 22026$ (fixed).

So the bias of c is directly determined by the sign and magnitude of δ . Since $\sum_{i \notin \{a,b\}} \alpha_i v_i \approx 0$ (negligible), $c \approx \alpha_a v_a + \alpha_b v_b$ with $\alpha_a + \alpha_b \approx 1$. The weights are:

$$\alpha_a = \frac{\exp(10 + 10\delta)}{\exp(10 + 10\delta) + \exp(10) + (n - 2)}, \quad \alpha_b = \frac{\exp(10)}{\exp(10 + 10\delta) + \exp(10) + (n - 2)}$$

- When $\delta > 0$: $10 + 10\delta > 10 \implies \exp(10 + 10\delta) > \exp(10) \implies \alpha_a > \alpha_b$. If $\delta \gg 0$ (e.g., $\delta = 0.5$), $\exp(10 + 10\delta) = \exp(15) \approx 3269017 \gg \exp(10)$, so $\alpha_a \approx 1$, $c \approx v_a$.

- When $\delta < 0$: If $\delta \ll 0$, $\alpha_a \approx 0$, $c \approx v_b$. (Calculation omitted)

- When $\delta = 0$: $\alpha_a = \alpha_b \approx 0.5$, $c \approx 0.5(v_a + v_b)$.

- For continuous $\delta \sim \mathcal{N}(0, 0.5)$, $\Pr(\delta = 0) = 0$, so this balanced state is negligible.

Conclusion: Unstable Swing of c . Since $\delta \sim \mathcal{N}(0, 0.5)$, $\Pr(\delta > 0) = \Pr(\delta < 0) = 0.5$, meaning c has an equal chance to approach v_a or v_b . This is a sharp contrast to Part (i) (all $\Sigma_i = \alpha I$), where $\delta \approx 0$ (negligible fluctuation), $\alpha_a \approx \alpha_b \approx 0.5$, and c stably remains the average of v_a and v_b . Here, the extra variance in Σ_a amplifies δ , breaking the balanced combination and causing c to swing randomly between v_a and v_b .

(d)

i This means divide $(v_a + v_b)$ into two different attention copies. Just as the previous question and let $q_1 = t\mu_a, q_2 = t\mu_b$ (where $t \gg 1$ is a large positive scalar, like $t = 5, 10, \dots$). Thus, $\alpha_a^{(1)} \approx 1$, $c_1 \approx v_a$ and $\alpha_b^{(2)} \approx 1$, $c_2 \approx v_b$.

The final output is $c = \frac{1}{2}(c_1 + c_2) \approx \frac{1}{2}(v_a + v_b)$.

ii Using $q_1 = t\mu_a$ and $q_2 = t\mu_b$ from part (i), and the fluctuation parameter $\delta \sim \mathcal{N}(0, 0.5)$ (describing $k_a^\top \mu_a = 1 + \delta$):

- $q_1 = t\mu_a$: The dot product $k_a^\top q_1 = t(1 + \delta)$. For $i \neq a$, $k_i^\top q_1 \approx 0 \implies \exp(k_i^\top q_1) = 1$. With $t \gg 1$, $\exp(t(1 + \delta)) \gg n - 1$ even for moderate δ , so:

$$\alpha_a^{(1)} \approx \frac{\exp(t(1 + \delta))}{\exp(t(1 + \delta)) + (n - 1)} \approx 1$$

Thus, $c_1 \approx v_a$ (nearly invariant to δ , as q_1 locks attention on v_a).

- $q_2 = t\mu_b$: $k_b \sim \mathcal{N}(\mu_b, \alpha I)$ (no extra variance) gives $k_b^\top q_2 \approx t$, so:

$$\alpha_b^{(2)} \approx \frac{\exp(t)}{\exp(t) + (n - 1)} \approx 1$$

Thus, $c_2 \approx v_b$ (stable, unaffected by δ).

- Combined output is

$$c = \frac{1}{2}(c_1 + c_2) \approx \frac{1}{2}(v_a + v_b)$$

This design isolates targets via dedicated queries: q_1 robustly captures v_a despite δ , and q_2 stably captures v_b , keeping c close to the desired average.

2 Pretrained Transformer models and knowledge access

(d)

- Correct: 13.0 out of 500.0: 2.6%

- London baseline accuracy: 25.0/500.0 (5.00%)

(f)

- Correct: 103.0 out of 500.0: 20.599999999999998%

(g)

- Correct: 73.0 out of 500.0: 14.6%

- Time Complexity Comparison of Transformer Models

Model Type	Time Complexity Expression	Core Logic Interpretation
Vanilla Transformer Model	$O(L\ell^2)$	All L layers use self-attention on input sequence length ℓ ; complexity grows with ℓ^2 .
Perceiver Model	$O(dm + Lm^2)$	Cross-attention contributes $O(dm)$ (d =input dim, m =bottleneck dim); L layers use self-attention on m . $m \ll \ell$ reduces computation drastically.

3 Considerations in pretrained knowledge

- (a) The pretrained vanilla model benefits from general linguistic and world knowledge learned during pretraining from 'Wikipedia.text'. It acquires patterns like name-to-region associations and contextual relationships between names and geographic terms. In contrast, the non-pretrained model starts with random weights and lacks this foundational knowledge, it cannot learn meaningful name-birthplace mappings from the limited fine-tuning data alone, resulting in accuracy below 10%.
- (b) This fact often called "hallucination" will raise critical issues for user-facing applications. Below are two distinct concerns with examples:
 - User-facing systems (e.g., historical research, genealogy tools) depend on accuracy. If the model hallucinates false birthplaces (e.g., "Nelson Mandela was born in Kenya"), users may unknowingly use this misinformation for academic or family history work. Unlike humans, the model provides no evidence for outputs, making independent verification impossible.
 - Everyday apps (e.g., cultural education tools for name origins) lose user trust due to consistent hallucinations. For example, if "Emma" is alternately linked to France and Germany, users will recognize unreliability over time and abandon the app—undermining its goal of delivering accurate cultural/geographic information.
- (c)
 - The model leverages statistical name patterns learned during pretraining to generalize: it identifies phonetic or morphological correlations between unseen names and the names it encountered in pretraining, then maps the unseen names to birthplaces associated with the pretrained names that share such correlations.
 - This pattern-based generalization leads to systematic bias and overgeneralization. For instance, the model may incorrectly map all names with similar roots to a single region (e.g., for whoever has a 'Li' in his/her name then all related to "Beijing, China")—even if the unseen name belongs to someone from a different country. In applications like global talent recruitment platforms (which use birthplace to tailor regional opportunities), this bias could misclassify candidates, denying them relevant opportunities based on unfounded name-region assumptions.